

YUN-LIANG ZHENG

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<https://al-yl.github.io/>

AI-YL



RESEARCH INTEREST

- Galaxy Formation and Its Connection to their host Galaxy Groups
- Intracluster/Intragroup Medium in Galaxy Group Systems
- X-ray Properties of Galaxy Groups and the Contributions of Different Components to X-ray Luminosity
- The Baryonic Budget for Galaxy Groups/Clusters Spanning Different Halo Mass Ranges

PROJECTS

Project 1

Hot Gas Properties and Evolution in Galaxy Clusters/Groups

2022 – Ongoing

- Measuring the X-ray Luminosities for DESI groups using eROSITA (eFEDS + eRASS:1).
- Exploring the correlation between X-ray and multi-waveband properties of central galaxies.
- Exploring the impact of different components of galaxy groups/clusters through stacked X-ray spectral properties.
- Tracing the history of hot gas within galaxy clusters based on Cosmological Hydrodynamic Simulations.

Project 2

Galaxy groups where the central galaxies are merging or are about to merge

2022 – Ongoing

- The properties of compact groups of galaxies and the properties of their galaxy members.
- The role of gas in different phases during the evolution of galaxy groups/clusters with different observational properties.
- The relationship between the assembly history and AGN feedback in galaxy groups/clusters with various observational properties.

EXPERIENCE

PostDoc

Shanghai Jiao Tong University

Feb 2022 – Ongoing Shanghai

Research Assistant (Intern)

Shanghai Astronomical Observatory

July 2014 – June 2015 Shanghai

EDUCATION

Ph.D. in Astrophysics

Shanghai Astronomical Observatory

Sept 2015 – Nov 2021

Thesis title: Compact Groups of Galaxies and the Compactness of Galaxy Groups

Training Contract (during the PhD period)

University of Science and Technology of China

Sept 2015 – Aug 2016

BSc. in Applied Physics

Shanghai Normal University

Sept 2010 – Aug 2014

GRANTS AND FUNDING

Research on the X-Ray Properties of DESI Galaxy Groups

China Postdoctoral Science Foundation Funded Project

Dec 2023 – Ongoing

STRENGTHS

Python (Main)

C

Fortran

R

CIGALE

ESO-MIDAS

EXSAS

eSASS

PUBLICATIONS

Journal Articles

- **Zheng, Yun-Liang**, Yang, X., Liu, T., Chen, S., Bulbul, E., Liu, A., Zhang, Y., Li, D., Kang, X., Gu, Y., Wang, Y., Li, Q., & Wang, J. (2025). Unveiling the Sources of X-ray Luminosity in DESI Galaxy Groups: Insights from the SRG/eROSITA All-Sky Survey. *submit to ApJ*.
 - Li, D., Fang, T., Ge, C., Liu, T., He, L., Li, Z., Nicastro, F., Yang, X., Zhang, X., & **Zheng, Yun-Liang**. (2024). Robust detection of hot intragroup medium in optically selected, poor galaxy groups by eROSITA. *arXiv e-prints*, arXiv:2412.01261. doi:10.48550/arXiv.2412.01261. arXiv: 2412.01261 [astro-ph.GA]
 - Wang, Y., Yang, X., Gu, Y., Xu, X., Xu, H., Wang, Y., Katsianis, A., Han, J., He, M., **Zheng, Yunliang**, Li, Q., Wang, Y., Hong, W., Wang, J., Tan, Z., Zou, H., Lange, J. U., Hahn, C., Behroozi, P., Aguilar, J. N., Ahlen, S., Brooks, D., Claybaugh, T., Cole, S., de la Macorra, A., Dey, B., Doel, P., Forero-Romero, J. E., Honscheid, K., Kehoe, R., Kisner, T., Lambert, A., Manera, M., Meisner, A., Miquel, R., Moustakas, J., Nie, J., Poppett, C., Rezaie, M., Rossi, G., Sanchez, E., Schubnell, M., Tarlé, G., Weaver, B. A., & Zhou, Z. (2024). Measuring the Conditional Luminosity and Stellar Mass Functions of Galaxies by Combining the Dark Energy Spectroscopic Instrument Legacy Imaging Surveys Data Release 9, Survey Validation 3, and Year 1 Data., 971(1), 119. doi:10.3847/1538-4357/ad5294. arXiv: 2312.17459 [astro-ph.GA]
 - **Zheng, Yun-Liang**, Yang, X., He, M., Shen, S.-Y., Li, Q., & Li, X. (2023). Measuring the X-ray luminosities of DESI groups from eROSITA Final Equatorial-Depth Survey - I. X-ray luminosity-halo mass scaling relation., 523(4), 4909–4922. doi:10.1093/mnras/stad1684. arXiv: 2306.02594 [astro-ph.GA]
 - **Zheng, Yun-Liang**, Shen, S.-Y., & Feng, S. (2022). The Compactness of Galaxy Groups in the Sloan Digital Sky Survey., 926(2), 119. doi:10.3847/1538-4357/ac43ba. arXiv: 2112.07871 [astro-ph.GA]
 - **Zheng, Yun-Liang**, & Shen, S.-Y. (2021). Compact Groups of Galaxies in Sloan Digital Sky Survey and LAMOST Spectral Survey. II. Dynamical Properties of Isolated and Embedded Groups., 911(2), 105. doi:10.3847/1538-4357/abeaa2. arXiv: 2102.12804 [astro-ph.GA]
 - **Zheng, Yun-Liang**, & Shen, S.-Y. (2020). Compact Groups of Galaxies in Sloan Digital Sky Survey and LAMOST Spectral Survey. I. The Catalogs., 246(1), 12. doi:10.3847/1538-4365/ab5c26. arXiv: 1911.11478 [astro-ph.GA]
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